INFOE 3 (Lab 1)
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BSINFO 3C

1. When you successfully used the payload to overwrite the secret_key variable, which pillar of the C.I.A. triad did you primarily violate? Explain why.

Answer:

- When the payload successfully overwrote the secret_key variable, the **Integrity** pillar of the CIA triad was primarily violated because sensitive data was modified without authorization, even though access to the system was denied.

2. **What specific vulnerability category did you exploit when you entered a string longer than 8 characters?**

Answer:

- The specific vulnerability category exploited was a **buffer overflow**, also known as an **out-of-bounds write**, which occurred because the program accepted input longer than the allocated buffer without performing proper bounds checking.

3. Why is your solution (#4 The Fix) more effective than simply "asking users to be honest"?

Answer:

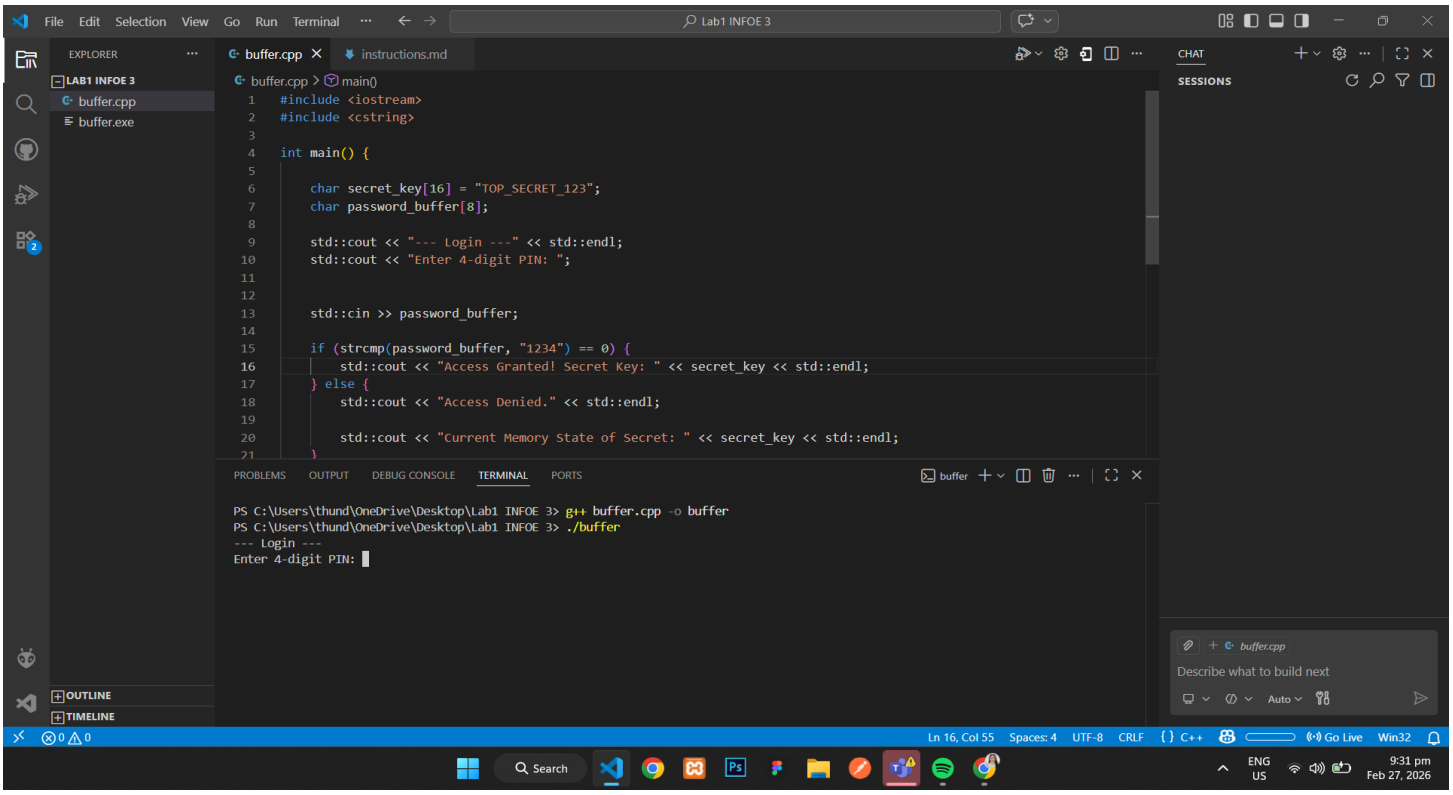
- The solution is more effective than asking users to be honest because security must be enforced through technical controls, and attackers cannot be relied upon to follow rules, warnings, or expected behavior.

4. **Imagine this was running the control system for a local power grid. Who might want to exploit this vulnerability, and what would be the real-world consequence if the secret key was overwritten by a professional cyber-attacker?**

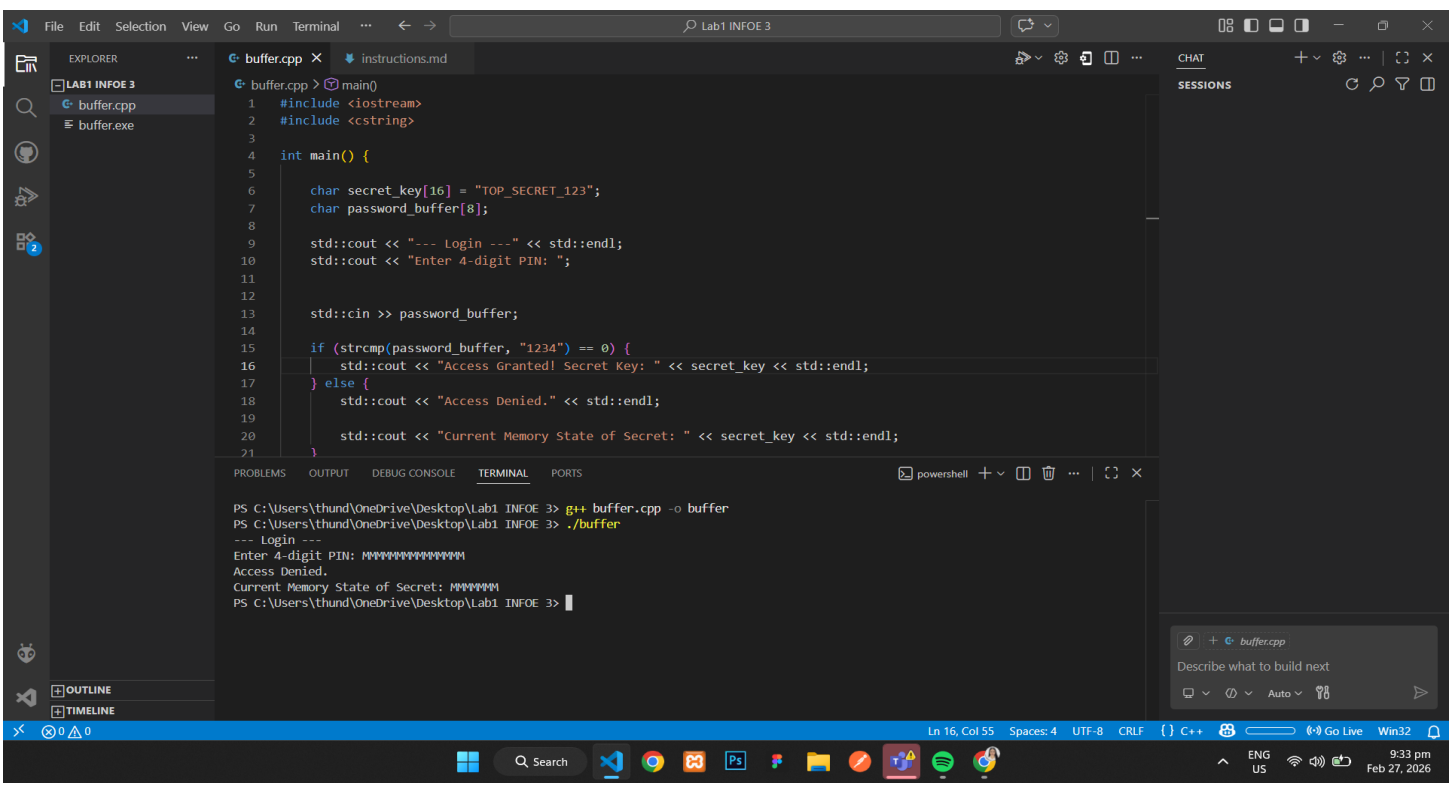
Answer:

- If this program were running a local power grid control system, attackers such as nationstate actors, cybercriminals, hacktivists, or malicious insiders might attempt to exploit the vulnerability, and if a professional cyber-attacker overwrote the secret_key controlling the power switch, it could result in power outages, disruption of critical infrastructure, economic damage, and serious risks to public safety.

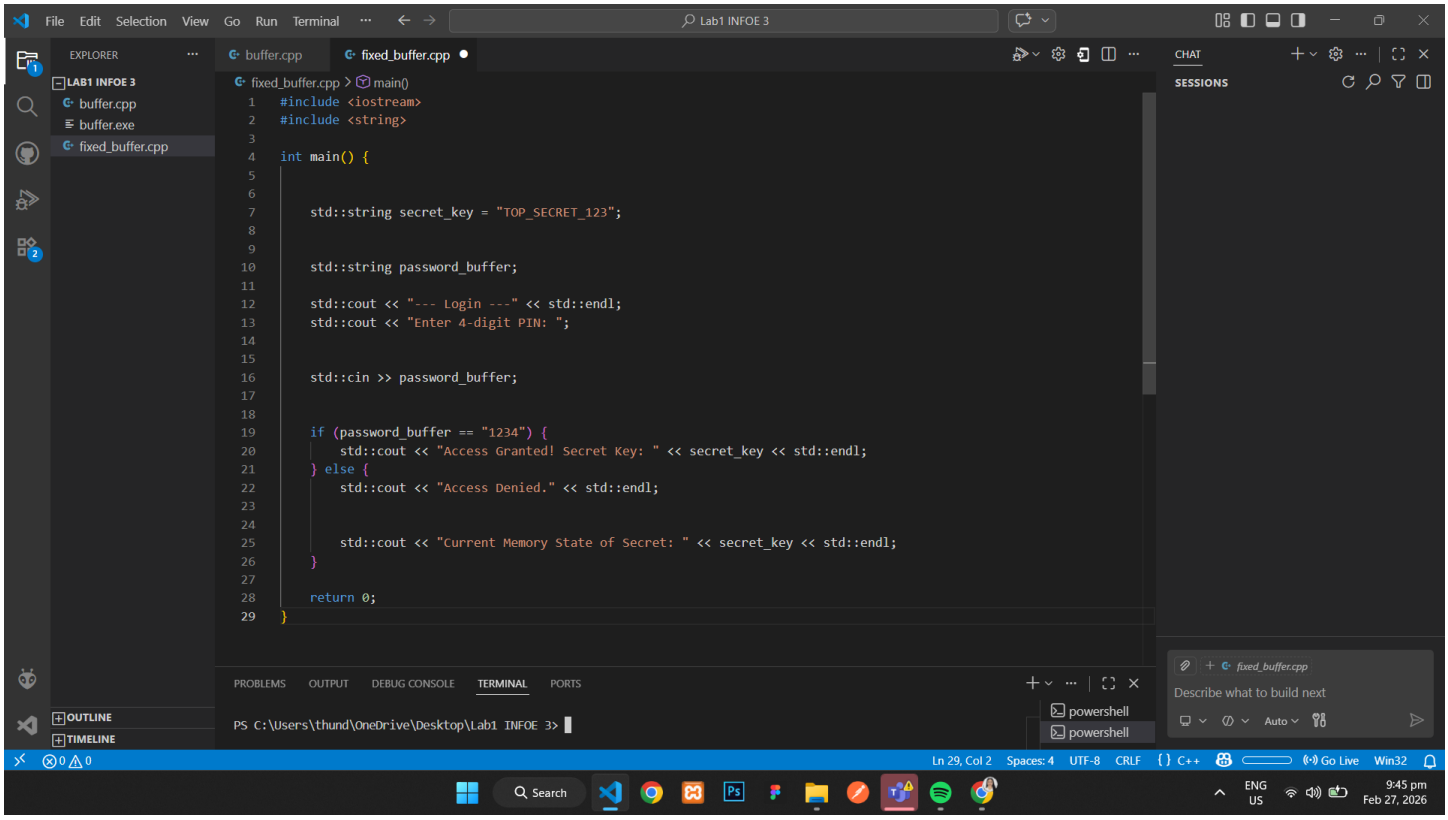
SCREENSHOOTS:



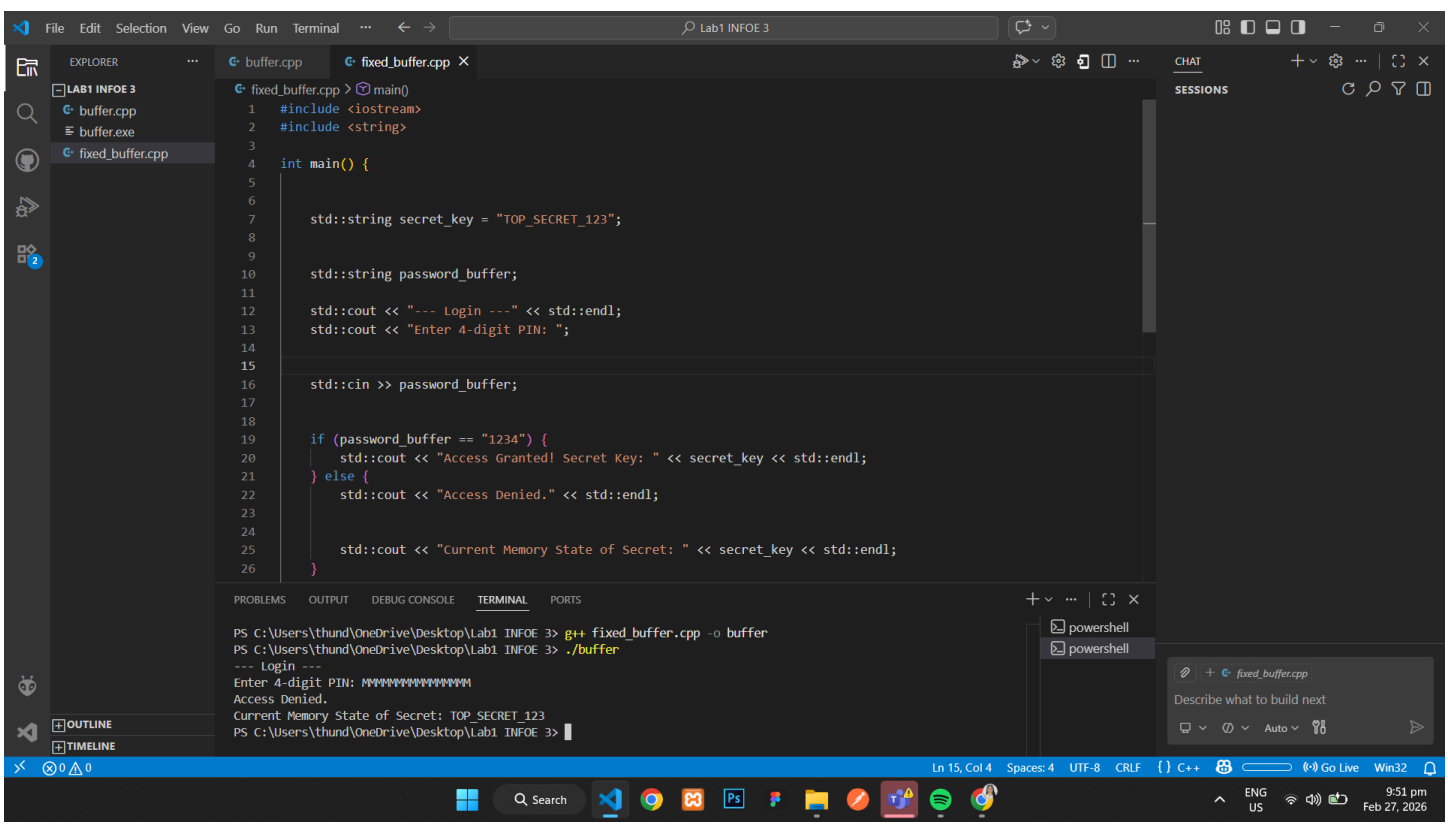
- This screenshot displays the initial configuration of the **buffer.cpp** program in **Visual Studio Code**.



- It illustrates how the buffer overflow vulnerability occurs during execution.



- This image presents the revised version, **fixed_buffer.cpp**, which applies safer memory handling techniques.



- The output verifies that the implemented fix successfully blocks the earlier exploit.